

Pedro Madruga

Principal AI Developer & Technical Lead | 12+ Years in Denmark

📍 Copenhagen ✉ hello@pedromadruga.com pedromadruga.com LinkedIn GitHub

Principal AI Developer with 6+ years in Data Science/ML and 10+ years in Software Engineering. Co-creator and AI lead of KAILA, a GenAI-powered legal assistant managing 3.5M documents, generating millions in revenue. Expert in Generative AI, RAG/DeepSearch, and NLP. Programming since the 90s, Linux user since 1999.

PROFESSIONAL EXPERIENCE

Principal AI Developer — Karnov Group AB

Oct 2024 - Present · Copenhagen

- Lead 5 AI Scientists as Technical AI Lead for Nordics on KAILA - GenAI legal assistant with 50+ stakeholders
- Scaled to 3.5M documents, generating millions in revenue, featured in major Danish newspapers
- Drive AI research using RAG, Agentic DeepSearch, and state-of-the-art information retrieval

Senior AI Developer — Karnov Group AB

Oct 2022 - Dec 2024 · Copenhagen

- Created KAILA from inception, scaling from 5k to 3.5M documents in 6 months
- Grew AI team from 1 to 5 developers, architected embeddings pipeline and cloud infrastructure
- Analyzed millions of search queries and developed multiple PoCs for business stakeholders

Senior Data Scientist — GoSimplo

Nov 2021 - Sep 2022 · Copenhagen

- Built deep learning models (Keras/TensorFlow) for financial prediction used by 100s of users
- Developed time-series models (XGBoost, ARIMA) deployed via Docker on Azure
- Processed large-scale data with Azure Synapse/SQL, delivered Streamlit dashboards

Data Scientist & Founder — JohnConnor.ai ApS

Jun 2020 - Nov 2021 · Copenhagen

- Founded AI/ML consultancy serving GrantCompass and Karnov Group
- Enhanced recommendation systems using AWS Personalize, built data warehouses

Data Scientist — Karnov Group Denmark

Jul 2019 - May 2020 · Copenhagen

- Enhanced Danish document recommendations using spaCy, Prodigy, content-based systems
- Achieved 93% reduction in response times (30s to 2s) enabling project success

Senior Software Engineer — Karnov Group Denmark

Apr 2017 - Mar 2020 · Copenhagen

Developed full-stack applications (React/Node.js), deployed on GCP/Kubernetes. Built Karnov Business Optimizer.

Founder & Software Engineer — Occam.ooo

Nov 2016 - Mar 2017 · Copenhagen

Technical lead for international projects. Delivered mobile apps for clients in Germany, Denmark, USA.

Software Engineer — colego.dk / Unwire

Feb 2014 - Oct 2016 · Copenhagen

Frontend developer and team lead. Developed Nordea payment apps, Android/iOS hybrid apps, re-implemented CMS.

Data Analyst — Copenhagenize Design Co.

Nov 2012 - Aug 2013 · Copenhagen

Analyzed Copenhagen cyclists' behavior, publication featured in Business Insider. Contributed to Copenhagenize Index.

EDUCATION

Machine Learning Certificate
DTU - Technical University of Denmark
2019

BSc Computer & Software Engineering
Instituto Superior de Engenharia de Lisboa
2003 - 2005

MSc + BSc Environmental Engineering
Universidade Nova de Lisboa
2006 - 2012

TECHNICAL SKILLS

AI/ML & Data Science

Generative AI & LLMs
RAG & DeepSearch
Information Retrieval
NLP & Transformers
Deep Learning
PyTorch & TensorFlow
Time Series Analysis

Programming & Tools

Python & JavaScript
SQL & NoSQL
Node.js & React.js
LangChain & PydanticAI
Docker & Kubernetes
spaCy & Prodigy
Git & Version Control

Cloud & Infrastructure

Google Cloud Platform
Microsoft Azure
AWS Services
PostgreSQL & MongoDB
CI/CD Pipelines
Monitoring & Logging
API Development

LANGUAGES

English: Native/Bilingual **Portuguese:** Native **Spanish:** Professional **Danish:** Elementary **French:** Limited Working

KEY ACHIEVEMENTS & RECOGNITION

- Co-created and scaled KAILA to 3.5M+ documents, generating millions in revenue with 50+ stakeholders
- Featured in major Danish newspapers for AI innovation in legal tech
- Published research on cyclist behavior featured in Business Insider
- Achieved 93% performance optimization enabling critical 2-year project success
- Founded successful AI/ML consultancy serving enterprise clients across Europe
- Built AI solutions adopted by top legal firms across Denmark and Sweden

LEADERSHIP & BUSINESS IMPACT

- Led cross-functional teams of 5+ AI scientists and 50+ stakeholders
- Drove product strategy from inception to multi-million revenue generation
- Established knowledge-sharing culture through workshops and technical sessions
- Successfully bridged technical innovation with business value delivery

ADDITIONAL INFORMATION

Open Source: Contributor to Danish NLP models and open-source AI projects
Certifications: SQL for Data Science, multiple cloud certifications
Interests: Supporting Danish language AI models, sustainable technology, urban mobility analytics